

Reminder per scope: this is a written specification artifact only — no real API is called, no real vendor is contacted. It's written as a hand-off document an engineering team could build against.

API Requirements & Data-Contract Specification

Third-Party KYC Vendor Integration (Hypothetical)

1. Purpose

This document specifies the data contract between our onboarding system and a hypothetical third-party KYC/identity-verification vendor (referred to generically as "Vendor" throughout, since no real vendor is integrated). It is intended as an engineering hand-off artifact, not a description of a live integration.

2. Field Mapping Table

Internal Field	Vendor Field (hypothetical)	Type	Notes
user_id	client_reference_id	string	Our internal ID, passed through unchanged so Vendor's response can be matched back to our record
country	nationality_country_code	string (ISO 3166-1 alpha-2)	Must be converted from our internal country name to ISO code before sending
id_document_image	document_front_image	base64 string	Not generated in this project — placeholder field for a real document upload
selfie_image	liveness_check_image	base64 string	Same as above — placeholder, no real image processing occurs
id_document_expiry_date	document_expiry	ISO 8601 date	Validated internally per FR-010 <i>before</i> this field is even sent to Vendor — we do not rely on

Internal Field	Vendor Field (hypothetical)	Type	Notes
			Vendor to catch expired documents
(computed internally) verification_match_rate	match_confidence_score	float (0.0-1.0)	Inbound field — Vendor returns this in their response; it is not something we send. Internally we rescale Vendor's 0.0-1.0 to our 0-100 scale: $\text{verification_match_rate} = \text{match_confidence_score} * 100$
(computed internally) PEP/sanctions flag	watchlist_match	boolean	Inbound field — Vendor's response. Maps directly to our is_pep/sanctions flag used in FR-005

3. Request Structure (hypothetical)

json

POST /v1/verify

```
{
  "client_reference_id": "U00001",
  "nationality_country_code": "IN",
  "document_front_image": "<base64>",
  "liveness_check_image": "<base64>",
  "document_expiry": "2027-03-15"
}
```

4. Response Structure (hypothetical)

```
json
{
  "client_reference_id": "U00001",
  "match_confidence_score": 0.87,
  "watchlist_match": false,
  "status": "completed",
  "vendor_reference_id": "VND-9928374"
}
```

5. Error-Handling Rules

HTTP Status	Vendor Meaning	Our System's Required Behavior
200	Success	Proceed to FR-001 risk score calculation using returned match_confidence_score and watchlist_match
400	Malformed request (e.g., invalid image format)	Do not retry automatically. Log error with client_reference_id. Show applicant a generic "we couldn't process your document, please try again" message — never expose the raw vendor error
422	Unprocessable entity — e.g., image quality too low for Vendor to assess, or document type unsupported	Treat as equivalent to an internal verification failure under FR-009 : prompt applicant to re-upload, counted against the same 2-attempt cap. User-facing message: <i>"We couldn't clearly read your document. Please retake the photo in good lighting and try again."</i> Do not surface the raw 422 or any vendor-specific language
429	Rate limited by Vendor	Queue and retry with exponential backoff (not user-facing — applicant should not see a delay-specific error unless backoff exceeds a defined timeout, at which point fall back to FR-013 's pending_verification state)

HTTP Status	Vendor Meaning	Our System's Required Behavior
500 / 502 / 503	Vendor-side outage	Do not default the applicant to a passing match rate or Low risk tier by omission — this directly implements FR-013 . Hold applicant in pending_verification, alert engineering on-call after N consecutive failures (N to be defined operationally — out of scope for this exercise)
Timeout (no response)	Network/Vendor non-response	Same as 500/502/503 — never silently proceed without a result

6. Explicit Cross-References to FRD

This document deliberately reuses, rather than redefines, the following FRD rules so the two documents stay consistent:

- 422 / failed-match handling → governed by **FR-009** (verification match rate failure), not a new rule
- Vendor outage handling → governed by **FR-013** (verification service unavailable), same logic, different trigger source
- Expired document → checked internally via **FR-010** *before* any data is sent to Vendor, since there's no reason to pay for/wait on a Vendor call for a document we already know is invalid

7. Out of Scope

- No real Vendor is named, contacted, or contracted
- No actual image upload, OCR, or biometric processing occurs anywhere in this project
- Rate limits, exact timeout durations, and retry counts are illustrative; finalizing them would require real Vendor SLA documentation that doesn't exist for this hypothetical integration